

DBMS

Assignment no 5

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Examples:

```
mysql> CREATE FUNCTION calc_grade(P INT) RETURNS text DETERMINISTIC
-> BEGIN
->     DECLARE g varchar(20);
->     IF P <= 1500 AND P >= 990 THEN
->         SET g = 'Distinction';
->     ELSEIF P <= 989 AND P >= 900 THEN
->         SET g = 'First Class';
->     ELSEIF P <= 899 AND P >= 800 THEN
->         SET g = 'Higher Second';
->     ELSE
->         SET g = 'FAIL';
->     END IF;
->     RETURN g;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> Create Procedure proc_grade()
->     Deterministic
->     BEGIN
->         Declare done INT Default False;
->         Declare nm varchar(20);
->         Declare total int(20);
->         Declare cur_grade cursor for select name , tot_marks from stud_marks;
->         Declare Continue Handler for NOT Found SET done = True;
->
->         Open cur_grade;
->
->         get_grade: LOOP
-> fetch cur_grade into nm,total;
->
->             IF done THEN
->                 Leave get_grade;
->             END IF;
->
->             Insert into result(name,grade)
->             values(nm,calc_grade(total));
->
->         END loop;
->
->         Close cur_grade;
->     END//
```

Query OK, 0 rows affected, 1 warning (0.01 sec)

-> DELIMITER ;

mysql> ALTER TABLE result MODIFY grade varchar(40);

Query OK, 0 rows affected (0.01 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE student (

-> roll_no INT PRIMARY KEY,

-> name VARCHAR(50)

->);

Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO student (roll_no, name) VALUES (1, 'Riya');

Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER \$\$

mysql> CREATE PROCEDURE duplicate_demo()

-> BEGIN

-> DECLARE EXIT HANDLER FOR 1062

-> BEGIN

-> SELECT 'Duplicate entry found!' AS Message;

-> END;

-> INSERT INTO student (roll_no, name) VALUES (1, 'Arya');

-> END \$\$

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> CALL duplicate_demo();

+-----+

| Message |

+-----+

| Duplicate entry found! |

+-----+

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Code:

```
mysql> create database ish;
```

Query OK, 1 row affected (0.02 sec)

```
mysql> use ish;
```

Database changed

```
mysql> create table stud_marks(roll_no int(5) not null primary key, name varchar(20), tot_marks int(5));
```

Query OK, 0 rows affected, 2 warnings (0.03 sec)

```
mysql> insert into stud_marks(roll_no,name,tot_marks)
```

```
-> values(1,"Isha",1200),
```

```
-> (2,"Ash" , 990),
```

```
-> (3, "Jah",800),
```

```
-> (4, "Gayatri",500),
```

```
-> (5, "Ira", 1250);
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> create table result(roll_no int(3) not null primary key,name varchar(20),grade varchar(25));
```

Query OK, 0 rows affected, 1 warning (0.03 sec)

```
mysql> DELIMITER $$
```

```
mysql> create function cal_grade(p int)
```

```
-> returns text
```

```
-> deterministic
```

```
-> begin
```

```
-> declare g varchar(20);
```

```
-> if p<=1500 and p>=990 then
```

```
-> set g="Distinction";
```

```
-> elseif p<=989 and p>=900 then
```

```
-> set g="First Class";
```

```
-> elseif p<=899 and p>=825 then
```

```
-> set g="Higher Second Class";
```

```
-> else
```

```
-> set g="Fail";
```

```
-> end if;
```

```
-> return g;
```

```
-> end $$
```

Query OK, 0 rows affected (0.01 sec)

```
DELIMITER ;
```

```
mysql> DELIMITER $$
```

```
mysql> create procedure proc_grade()
```

```

-> deterministic
-> begin
-> declare done int default false;
-> declare rno int(5);
-> declare nm varchar(20);
-> declare total int(20);
-> declare cur_grade cursor for select roll_no,name,tot_marks from stud_marks;
-> declare continue handler for not found set done = true;
-> open cur_grade ;
-> get_grade:loop
-> fetch cur_grade into rno,nm,total;
-> if done then
-> leave get_grade;
-> end if;
-> insert into result(roll_no,name,grade)
-> values(rno,nm,cal_grade(total));
-> end loop;
-> close cur_grade;
-> end $$

```

Query OK, 0 rows affected, 2 warnings (0.02 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL proc_grade();
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> select * from result;
```

```

+-----+-----+-----+
| roll_no | name   | grade   |
+-----+-----+-----+
| 1 | Isha   | Distinction |
| 2 | Ash    | Distinction |
| 3 | Jah    | Fail       |
| 4 | Gayatri | Fail       |
| 5 | Ira    | Distinction |
+-----+-----+-----+

```

5 rows in set (0.00 sec)
